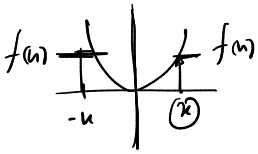
Lecture - 17

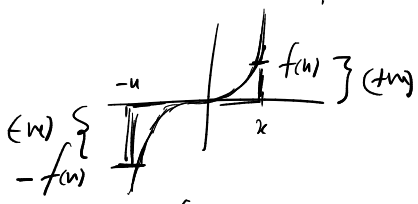
$f(x)$ is even, $\underbrace{f(-x) = f(x)}$



$f(x) = x^2$

symmetry w.r.t. y axis

$f(x)$ is odd, $\underbrace{f(-x) = -f(x)}$



$f(x) = x^3$

$$\underbrace{\int_{-L}^{L} \text{even}\, dx}_{\text{Area}} = \boxed{2\int_{0}^{L} \text{even}\, dx}$$

$$\boxed{\int_{-L}^{L} \text{odd}\, dx = 0}$$

Fourier series

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos\left(k \frac{x\pi}{L}\right) + \sum_{k=1}^{\infty} b_k \sin\left(k \frac{x\pi}{L}\right)$$

$$a_k = \frac{1}{L}\int_{-L}^{L} f(x) \cos k\left(\frac{x\pi}{L}\right) dx$$

$$b_k = \frac{1}{L}\int_{-L}^{L} f(x) \sin k\left(\frac{x\pi}{L}\right) dx$$

Domain $[-L, L]$

$f(x)$ even, $\rightarrow$ even combination

$\boxed{b_k = 0}$

$$a_k = \frac{1}{L}\int_{-L}^{L} \underbrace{\underbrace{f(x)}_{\text{even}} \underbrace{\cos k\left(\frac{x\pi}{L}\right)}_{\text{even}}}_{\text{even}} dx$$

$$= \frac{2}{L}\int_{0}^{L} f(x) \cos k\left(\frac{x\pi}{L}\right) dx$$

$f(x)$ odd, $\rightarrow$ odd combination

$\boxed{a_k = 0}$

$$b_k = \frac{1}{L}\int_{-L}^{L} \underbrace{f(x)}_{\text{odd}} \underbrace{\sin k\left(\frac{x\pi}{L}\right)}_{\text{odd}} dx$$

$$= \frac{2}{L}\int_{0}^{L} f(x) \sin k\left(\frac{x\pi}{L}\right) dx$$

even × even = even
odd × even = odd
odd × odd = even

$$\left.\begin{aligned} &f(x) = x \\ &f(-x) = -x = -f(x) \end{aligned}\right\} \text{odd}$$

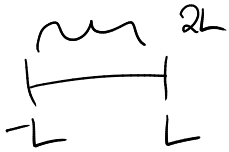
$$\left.\begin{aligned} &f(x) = |x| \\ &f(-x) = |x| = f(x) \end{aligned}\right\} \text{even}$$

$$\left.\begin{aligned} &\boxed{f(x) = \frac{1}{2}(\pi - x)} \\ &f(-x) = \frac{1}{2}(\pi + x) \end{aligned}\right\} \text{odd/even} \times$$

Example $f(x) = \frac{1}{2}(\pi - x)$, $\boxed{[0, 2\pi]}$ $\quad 2\pi = 2L \Rightarrow \boxed{L = \pi}$

$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos(kx) + \sum_{k=1}^{\infty} b_k \sin(kx)$$

$$k \times \frac{x\pi}{L} = k \times \frac{x\cancel{\pi}}{\cancel{\pi}} = kx$$



$$a_0 = \underbrace{\frac{1}{\pi}\int_0^{2\pi}}_{\text{Domain period}} f(x)\,dx = \underbrace{\frac{1}{2\pi}\int_0^{2\pi}(\pi - x)\,dx}_{} = 0$$

$$a_k = \frac{1}{2\pi}\int_0^{2\pi}(\pi - x)\cos(kx)\,dx =$$

	u	v
+	$\pi - x$	$\cos(kx)$
−	-1	$\frac{\sin(kx)}{k}$
	0	$-\frac{\cos(kx)}{k^2}$

$$= \frac{1}{2\pi}\left[(\pi - x)\frac{\sin(kx)}{k} - \frac{\cos(kx)}{k^2}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left[-\pi \underbrace{\frac{\sin(k2\pi)}{k}}_{=0} - \frac{\cos(k2\pi)}{k^2} - 0 + \frac{\cos 0}{k^2}\right]$$

$$= \frac{1}{2\pi}\left(-\frac{1}{k^2} + \frac{1}{k^2}\right)$$

$$= 0.$$

$$(e^{i\pi})^{2k} = \cos(2k\pi) + i\sin(2k\pi)$$

$$(-1)^{2k}$$

$$\left.\begin{aligned} 1 &= \cos(2k\pi) \\ 0 &= \sin(2k\pi) \end{aligned}\right\}$$

$$b_k = \frac{1}{2\pi}\int_0^{2\pi}(\pi - x)\sin(kx)\,dx \qquad -(\pi - 2\pi) = -\pi$$

	u	v
+	$\pi - x$	$\sin(kx)$
−	-1	$-\frac{\cos(kx)}{k}$
	0	$-\frac{\sin(kx)}{k}$

$$= \frac{1}{2\pi}\left[-(\pi - x)\frac{\cos(kx)}{k} - \frac{\sin(kx)}{k}\right]_0^{2\pi}$$

$$= \frac{1}{2\pi}\left(\pi\frac{\cos(k2\pi)}{k} - \frac{\sin(k2\pi)}{k} + \pi\frac{1}{k} + 0\right)$$

$$= \frac{1}{2\pi}\left(\frac{\pi}{k}+\frac{\pi}{k}\right) = \frac{1}{\cancel{2\pi}} \times \frac{\cancel{2\pi}}{k} = \boxed{\frac{1}{k}}$$

$$\underbrace{f(x)} = \sum_{k=1}^{\infty} \frac{1}{k}\sin(kx)$$

Find out $\boxed{1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\cdots = \boxed{\frac{\pi}{4}}}$ ✓ $\longrightarrow$ (A)

$$\boxed{\underbrace{f(x)} = \frac{1}{1}\sin(x) + \frac{1}{2}\sin(2x) + \cdots}$$

$$\boxed{\frac{1}{2}(\pi - x) = \sin(x) + \frac{1}{2}\sin(2x) + \cdots}$$

$$\boxed{\frac{\pi}{4} = \frac{1}{2}(\pi - x)}$$

$$\frac{\pi}{4} = \frac{\pi}{2} - \frac{x}{2} \Rightarrow \frac{\pi}{4} - \frac{\pi}{2} = -\frac{x}{2} \Rightarrow \frac{\pi - 2\pi}{4} = \frac{-x}{2} \Rightarrow \cancel{-}\frac{\pi}{4} = \cancel{-}\frac{x}{2}$$

$$\Rightarrow x = \pi/2$$

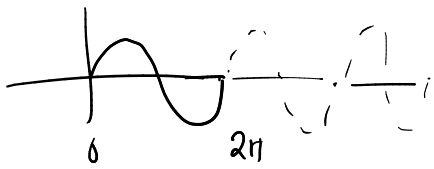
(A) insert $x = \pi$,

$$\sin\left(\frac{\pi}{2}\right) + \underbrace{\frac{1}{2}\sin(\pi)} + \frac{1}{3}\sin\left(\frac{3\pi}{2}\right) + \cdots = \frac{\pi}{4}$$

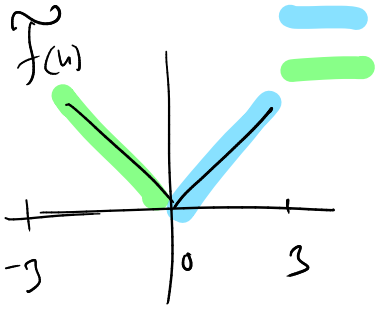
$$\boxed{1 + 0 - \frac{1}{3} + \cdots = \frac{\pi}{4}}$$ ✓

Half interval / sine or cosine series

$(-L, L)$ $\quad$ $\underbrace{(0, 2L)}$



If we are given half interval, $(0,L) \longrightarrow (-L,L)$



given
extension

Even

Fourier cosine series

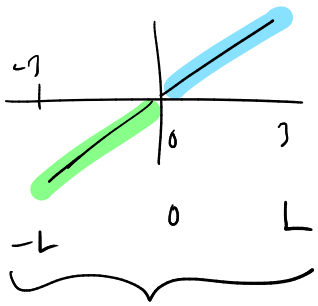
$$f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} a_k \cos k\left(\frac{x\pi}{L}\right)$$

$b_k = 0$

$$a_k = \frac{1}{L}\int_{-L}^{L} \tilde{f}(x) \cos k\left(\frac{x\pi}{L}\right) dx$$

Even Ext.

$$= \frac{2}{L}\int_{0}^{L} f(x) \cos k\left(\frac{x\pi}{L}\right) dx$$



$\boxed{\text{Odd}}$

Fourier sine series

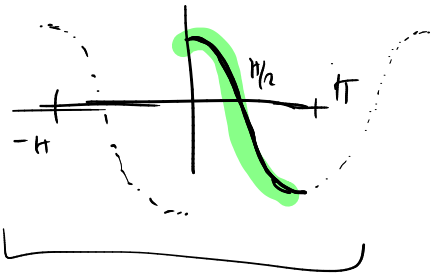
$$f(x) = \sum_{k=1}^{\infty} b_k \sin k\left(\frac{x\pi}{L}\right)$$

$a_k = 0$

$$b_k = \frac{2}{L}\int_{0}^{L} f(x) \sin k\left(\frac{x\pi}{L}\right) dx$$

Example

$f(x) = \cos(x)$ $(0,\pi)$ find the Fourier sine series



$$\boxed{f(x) = \sum_{k=1}^{\infty} b_k \sin(kx)}$$

$$\boxed{\int_0^L \text{ or } \int_{\text{given domain}}}$$

$$b_k = \frac{2}{\pi}\int_0^{\pi} f(x) \sin(kx)\, dx$$

$$b_k = \frac{2}{\pi}\int_0^{\pi} \cos(x) \sin(kx)\, dx$$

$$2\sin A\cos B = \boxed{\sin(A+B) - \sin(A-B)}$$

$$= \frac{1}{\pi}\int_0^{\pi} \underbrace{2\cos(x)\sin(kx)}\,dx$$

$$= \frac{1}{\pi}\int_0^{\pi}\left(\sin(k+1)x - \sin(k-1)x\right)dx$$

$$= \frac{1}{\pi}\int_0^{\pi}\sin(k+1)x\,dx - \frac{1}{\pi}\int_0^{\pi}\sin(k-1)x\,dx$$

$$= \frac{1}{\pi} - \frac{\cos(k+1)x}{k+1}\Bigg|_0^{\pi} + \frac{1}{\pi}\,\frac{\cos(k-1)x}{k-1}\Bigg|_0^{\pi}$$

$$= \frac{1}{\pi}\left[-\frac{\cos(k+1)\pi}{k+1} + \frac{1}{k+1} + \frac{\cos(k-1)\pi}{k-1} - \frac{1}{k-1}\right]$$

$$= \frac{1}{\pi}\left[-\frac{(-1)^{k+1}}{k+1} + \boxed{\frac{1}{k+1}} + \frac{(-1)^{k-1}}{k-1} - \frac{1}{k-1}\right]$$

$$= \frac{1}{\pi}\left[-\frac{(-1)^k\times(-1)^1}{k+1} + \frac{\cancel{k-1}-\cancel{k-1}}{k^2-1} + \frac{\frac{(-1)^k}{(-1)}}{k-1}\right]$$

$$= \frac{1}{\pi}\left[\frac{(-1)^k}{k+1} + \frac{-2}{k^2-1} - \frac{(-1)^k}{k-1}\right]$$

$$= \frac{1}{\pi}\left(\frac{\cancel{(-1)^k k} - (-1)^{k+1} - \cancel{(-1)^k k} - (-1)^k\times 1}{k^2-1} + \frac{-2}{k^2-1}\right)$$

$$= \frac{1}{\pi}\left(\cancel{\frac{(-1)^k - (-1)^k}{k^2-1}} + \frac{-2}{k^2-1}\right)$$

$$b_k = \frac{-2}{\pi(k^2-1)}$$

$$f(x) = \sum_{k=1}^{\infty}\frac{-2}{\pi(k^2-1)}\sin(kx)$$

$$\left(e^{i\pi}\right)^{k+1} = \cos(k+1)\pi + i\sin(k+1)\pi$$

$$\underbrace{e^{i\pi(k+1)}}$$

$$e^{i\pi k}\times e^{i\pi}$$

$$(-1)^k\times(-1)$$

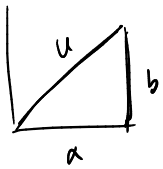
$$\underbrace{(-1)^{k+1}} = \underbrace{\cos(k+1)\pi} + i\sin(k+1)\pi$$

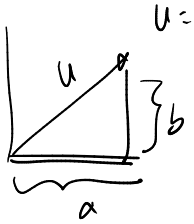
$$(-1)^{k+1} = \cos(k+1)\pi$$

$$0 = \sin(k+1)\pi$$

Parseval's identity

$$\boxed{u^2 = a^2 + b^2}$$





$$u = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$|u| = \sqrt{a^2 + b^2}$$

$$|u|^2 = a^2 + b^2$$

$$|u| = \sqrt{\langle u, u \rangle}$$

$$\boxed{|u|^2 = \langle u, u \rangle}$$

$$\boxed{\langle u, u \rangle = a^2 + b^2}$$

$$u = a\vec{e}_1 + b\vec{e}_2$$

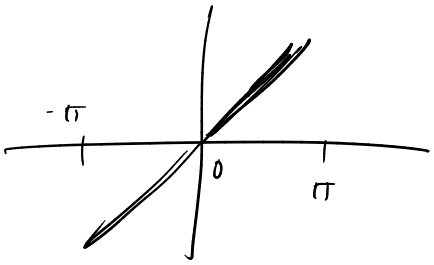
$$f(x) = \frac{a_0}{2} \times 1 + \sum_{k=1}^{\infty} a_k \cos\left(k \times \frac{x\pi}{L}\right) + \sum_{k=1}^{\infty} b_k \sin\left(k \frac{x\pi}{L}\right)$$

$$\langle f(x), f(x) \rangle = \frac{a_0^2}{2} + \sum_{k=1}^{\infty} a_k^2 + \sum_{k=1}^{\infty} b_k^2$$

$$\frac{1}{L} \int_{-L}^{L} f(x)^2 \, dx = \frac{a_0^2}{2} + \sum_{k=1}^{\infty} \left(a_k^2 + b_k^2\right)$$

Example

$$\boxed{f(x) = x}$$



OR,

$$\boxed{\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}}$$

$$f(x) = \sum_{k=1}^{\infty} \underbrace{-\frac{2(-1)^k}{k}}_{b_k} \sin(kx)$$

$$a_k = 0, \quad a_0 = 0$$

$$\langle \underbrace{f(x), f(x)} \rangle = \frac{a_0^2}{2} + \sum_{k=1}^{\infty} a_k^2 + b_k^2$$

$$\langle x, x \rangle = \sum_{k=1}^{\infty} \frac{4(-1)^{2k}}{k^2}$$

$$\underbrace{\frac{1}{\pi}\int_{-\pi}^{\pi} x^2\, dx} = \sum_{k=1}^{\infty} \frac{4}{k^2}$$

$$\frac{1}{\pi}\left(\frac{x^3}{3}\Big|_{-\pi}^{\pi}\right) = \sum_{k=1}^{\infty} \frac{4}{k^2}$$

$$\frac{1}{\pi}\left(\frac{\pi^3}{3} + \frac{\pi^3}{3}\right) = \sum_{k=1}^{\infty} \frac{4}{k^2}$$

$$\frac{\cancel{2}\pi^2}{\cancel{6}} = \sum_{k=1}^{\infty} \frac{\cancel{4}}{k^2}$$

$$\boxed{\frac{\pi^2}{6} = \sum_{k=1}^{\infty} \frac{1}{k^2}}$$